WORLDPRISM: 3D Consistency for Video World Models via Bidirectional Cross-Space Verification

Hengyu Liu[1,*], **Jiahao Lu**[2,*], **Ruijie Zhu**[3],
Sixiao Zheng[4,5], **Yuan Liu**[2], **Wenbo Hu**[6,†], **Ying Shan**[6]
[1]CUHK, [2]HKUST, [3]NTU, [4]FDU, [5]SII, [6]ARC Lab, Tencent IEG

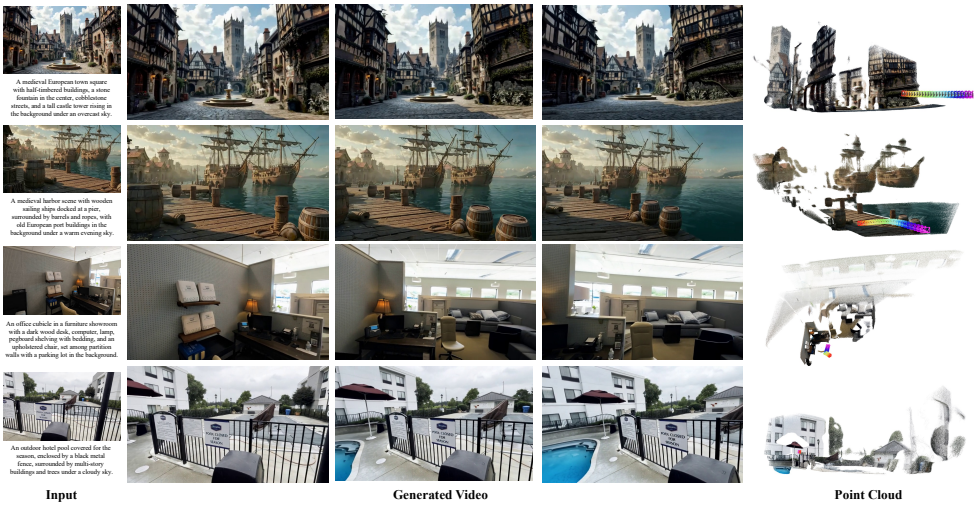

Figure 1: WORLDPRISM enables geometrically consistent world modeling for video generation via reinforcement learning. Given reference images, text prompts, and action sequences, it produces high-quality videos with strong 3D consistency. We visualize both the generated frames and their corresponding 3D reconstructions.

Abstract

Video world models aim to generate temporally coherent visual observations conditioned on actions, but often suffer from 3D geometric inconsistencies such as spatial drift, object deformation, and perspective violations, which limit their reliability in downstream tasks. We propose WORLDPRISM, a post-training framework that improves 3D consistency via reinforcement learning without modifying the base model architecture. The central challenge lies in designing a reliable reward signal. To address this, we introduce *Bidirectional Geometry-Flow Verification*, a bidirectional cross-space mechanism that evaluates consistency through two complementary paths: (i) lifting 2D optical flow into 3D to measure position-space agreement, and (ii) projecting reconstructed 3D geometry to match the point correspondence. By leveraging these two independent and complementary signals, this design mitigates the blind spots of unilateral 2D evaluation. We further incorporate hierarchical temporal evaluation to capture both local and long-range geometric consistency. Experiments on diverse datasets demonstrate that WORLDPRISM achieves state-of-the-art 3D consistency while preserving visual quality and generative diversity. Project page is here.

*Equal Contribution, † Corresponding Author.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

1 Introduction

Video world models [1, 2, 3, 4, 5, 6, 7] learn to autoregressively generate visual observations conditioned on actions, functioning as learned simulators that support downstream tasks such as planning, navigation, and manipulation. A critical quality axis for such models is *3D geometric consistency*, the requirement that a generated video faithfully depicts a single, coherent 3D world [8, 9, 10, 11]. Yet even models pretrained on internet-scale data still exhibit object deformation, spatial drift, and perspective violations under camera motion [12, 13, 14], directly undermining downstream decision-making: an embodied agent relying on spatially inconsistent predictions can produce erroneous plans.

Various strategies have been explored to improve this consistency. One line of work maintains explicit 3D memory during generation, storing per-frame scene representations [15, 16], accumulated point clouds [17], or pose-aware latent memory banks [18]. Camera-conditioned approaches [19, 20, 21] take a different route, encoding explicit camera trajectories such as Plücker embeddings or extrinsic matrices as spatiotemporal control signals injected into the denoising process. Geometric regularization methods [11] instead introduce 3D losses during training to encourage structural fidelity. While effective to varying degrees, these approaches either modify the model architecture and inference pipeline with substantial computational overhead [15, 17] or require large-scale 3D ground-truth data that is difficult to curate [11, 19], limiting their applicability as a general-purpose alignment mechanism for pretrained video world models [12, 22].

These limitations motivate post-training as an attractive alternative: rather than modifying the model architecture or curating extensive 3D supervision, one can align a pretrained video model toward better geometric consistency using reinforcement learning [23, 24, 25, 26, 27]. By preserving the original generative capacity while steering its outputs, this paradigm shifts the core difficulty from architectural design to reward design: *how to define and measure 3D consistency as a reliable training signal?* Existing approaches [12, 13] adopt a *unilateral* evaluation strategy that assesses geometric fidelity through a single forward projection into 2D. For example, VideoGPA [12] reprojects 3D structure to the image plane and measures RGB differences, while Kupyn et al. [13] evaluate epipolar constraints via the Sampson error [28]. However, optimizing solely on such 2D-space rewards introduces inherent blind spots. First, the irreversible 3D-to-2D projection only reveals visible regions, potentially hiding underlying geometric inconsistencies. Second, evaluating in RGB space often entangles true structural errors with view-dependent appearance variations [29]. Driven by these incomplete signals, the model's optimization is limited by what can be observed, imposing a ceiling on the 3D consistency achievable through post-training.

To address this challenge, we introduce WORLDPRISM, a post-training framework with *Bidirectional Geometry-Flow Verification (BGFV)* that aims to enhance the 3D consistency of current video world models. Specifically, it includes two independent and complementary streams: *Flow-Guided Geometry Verification (F2G)* and *Geometry-Induced Flow Verification (G2F)*. Given a video sequence generated by video world models, we utilize off-the-shelf geometry foundation models [30, 31] and an optical flow estimation model [32] to estimate 3D geometry and 2D flow. In *F2G*, we first use the estimated optical flow to find matching points between the source and target frames, then project them into 3D space to compute the 3D position error. In *G2F*, we first reproject the reconstructed 3D point clouds from the source frame to the target frame to produce geometry-induced 2D flow, then supervise it using the directly estimated optical flow. By spanning both spaces with independent signals, each stream covers the structural blind spots of the other: geometric violations invisible under one stream are exposed by the complementary one, as shown in Fig. 2. Moreover, since the two streams rely on *independent* and *orthogonal* external models, their mutual agreement on high error constitutes compelling evidence of genuine 3D violations rather than artifacts of any single perceptual backbone. WORLDPRISM further organizes this reward into a hierarchical temporal evaluation spanning local, structural, and global scales with curriculum-scheduled weighting, and integrates the resulting signal into a DiffusionNFT [33]-based reinforcement learning pipeline for effective post-training.

We validate WORLDPRISM on HunyuanVideo [4, 34] equipped with the WorldPlay [3] autoregressive action conditioning, where it yields substantial improvements in 3D geometric consistency while faithfully preserving visual quality and generative diversity. Our contributions are three-fold:

- We identify a fundamental bottleneck in existing post-training methods for 3D-consistent video world models: they rely on a 2D projection of a single evaluation space to assess

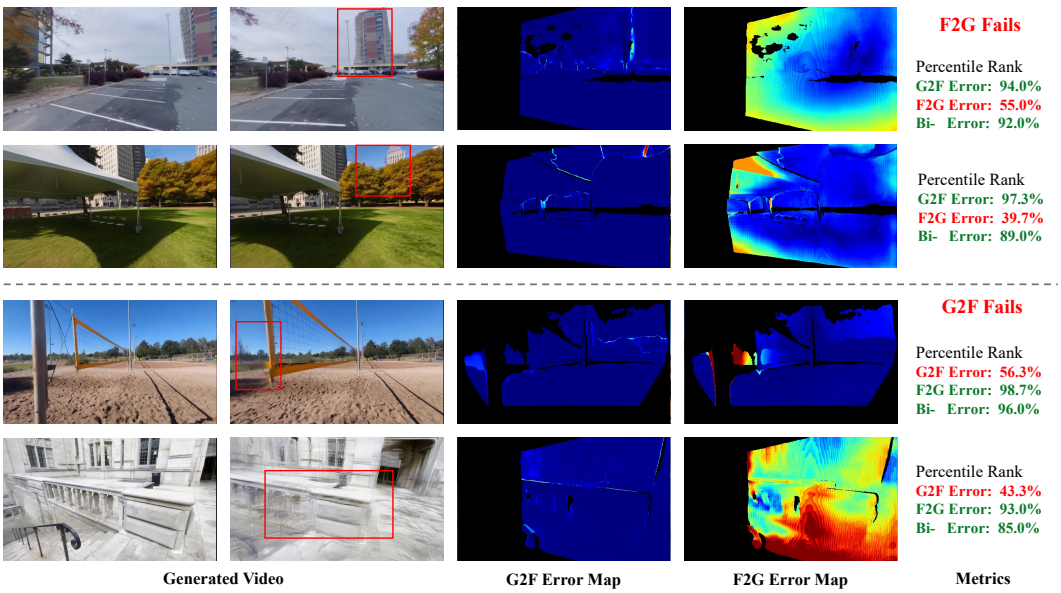

Figure 2: **Reward Robustness Case Study.** Failures in single-path verification are detected by our bidirectional reward. Percentile ranks of G2F and F2G errors show each frame pair's position among all processed pairs, where higher values indicate higher ranks.

consistency, thereby inheriting the systematic blind spots of 2D projections, and lack an independent self-correction mechanism.

- We design a *Bidirectional Geometry-Flow Verification* mechanism comprising two complementary streams that enable both Geometry-Induced Flow Verification and Flow-Guided Geometry Verification. The bidirectional design delivers a robust 3D consistency signal, minimizing inductive biases inherent in individual perception models.
- We propose WORLDPRISM, a complete post-training framework for video world models, and demonstrate that it can significantly improve 3D consistency on the HunyuanVideo model (using WorldPlay as the action model).

2 Related Work

3D Consistency in Video Generation. Approaches to improving the 3D geometric consistency of generated videos can be broadly categorized by *where* geometric constraints are imposed in the model lifecycle. *Architecture-level methods* integrate explicit 3D representations into the generation pipeline. DAS [16] introduces long-term 3D trajectories for controllable world modeling; Zhang et al. [15] maintain a persistent 3D scene updated via differentiable rendering; Wu et al. [17] accumulate a global point cloud as a geometric prior for frame synthesis; and WorldMem [18] employs a pose-indexed latent memory queried via cross-attention to retrieve prior geometry. *Camera-conditioned methods* [19, 20, 21] encode camera trajectories as explicit control signals. CameraCtrl [19] injects Plücker ray embeddings into temporal attention for 6-DoF control; CameraCtrl-II [20] extends this to dynamic scenes by disentangling camera and object motion; and Trajectory-Attention [21] introduces a trajectory-aware attention branch to complement temporal attention, enabling coordinated motion control and content generation. *Geometric regularization* [11] incorporates depth and epipolar consistency losses during training to improve structural fidelity without modifying the inference architecture. While effective, these approaches impose constraints *before or during* generation, either requiring architectural modifications with added complexity or relying on large-scale 3D supervision, making them difficult to apply to pretrained models. This motivates post-training as a complementary paradigm that aligns pretrained models *after* generation via reward-driven optimization.

Post-Training for Video World Models. Post-training through reinforcement learning has emerged as a flexible paradigm for aligning pretrained diffusion models without architectural modification.

Early work applies RLHF-style optimization to image diffusion models for text alignment and aesthetic quality [24, 27, 23]. More recent methods extend this to flow matching video models: FlowGRPO [25] and DanceGRPO [26] adapt group relative policy optimization to the continuous denoising setting, while DiffusionNFT [33] introduces a negative-aware fine-tuning strategy that explicitly leverages low-reward samples as contrastive signals. WorldCompass [35] applies DiffusionNFT to video world models with action-following and visual quality rewards. As established in these works, the post-training paradigm shifts the core difficulty from architectural design to *reward design*, and the quality of the reward signal directly determines the ceiling of achievable alignment.

Reward Design for 3D Consistency. When post-training targets 3D consistency specifically, the central challenge becomes how to measure geometric fidelity reliably enough to serve as a training signal. Existing approaches predominantly adopt a *unilateral* evaluation strategy: they assess consistency through a single verification direction within a single evaluation space. WorldScore [8] applies DROID-SLAM [36] to compute dense reprojection error; VideoGPA [12] and GeCo [37] leverage geometry foundation models [38, 39] for depth-based reprojection; Kupyn et al. [13] evaluate epipolar constraints with Sampson error [28]; and VIGOR [40] computes cross-frame reprojection error against independently tracked correspondences. Despite differences in signal sources, these methods share a common structural limitation: committing to a single projection direction inherits the systematic blind spots of that particular mapping, and operating in a single evaluation space provides no independent mechanism to distinguish model noise from genuine geometric violations. This places an inherent ceiling on the 3D consistency achievable through such rewards. WORLDPRISM lifts this ceiling through two complementary paths operating in opposite directions, one lifting 2D correspondences into 3D space, the other projecting 3D geometry into 2D space, cross-check geometry-flow consistency in both directions. Because the paths rely on external models with orthogonal failure modes, their mutual agreement on high error constitutes compelling evidence of genuine 3D violations rather than artifacts of any single perceptual backbone. While WORLDPRISM verifies 3D consistency in both the 3D geometry space [30, 41] and the 2D flow space [32, 42], rather than relying on a single evaluation space.

3 Method: WORLDPRISM

3.1 Problem Formulation

Given an image condition c, a sequence of discrete actions $a_{1:N}$, and a target clip index n to optimize, we first autoregressively generate the preceding $n-1$ clips as shared prefix context $x_{1:n-1}$, then produce G candidate rollouts for the n-th clip:

$$x_{1:n-1} = \pi_\theta(\cdot \mid a_{1:n-1}, c), \quad x_n^{(i)} = \pi_\theta(\cdot \mid x_{1:n-1}, a_n, c), \quad i = 1, \dots, G. \tag{1}$$

Each clip is generated by a flow-matching diffusion transformer operating in latent space. Let $\mathbf{z}_0 = \mathcal{E}(x_n)$ denote the encoded latent of a target clip. The forward process constructs noisy latents via linear interpolation:

$$\mathbf{z}_\sigma = (1-\sigma)\,\mathbf{z}_0 + \sigma\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2}$$

where $\sigma \in [0, 1]$ follows a shifted cosine schedule. The model $v_\theta(\mathbf{z}_\sigma, \sigma, x_{1:n-1}, a_n, c)$ learns to predict the velocity field that denoises $\mathbf{z}_\sigma$ toward $\mathbf{z}_0$.

Each candidate is scored by a reward function $R(x_n^{(i)}, a_n, c)$ that quantifies the 3D geometric consistency of the generated clip. The central challenge is the design of R: a reward that reliably captures 3D violations without the systematic blind spots of unilateral 2D evaluation. We address this through Bidirectional Geometry-Flow Verification (§3.2), a bilateral cross-space reward that cross-checks optical flow against 3D reconstruction through two complementary verification paths.

3.2 Bidirectional Geometry-Flow Verification

Rather than committing to a single projection direction, we cross-check 3D consistency through two independently sourced geometric signals, dense optical flow $\mathbf{f}_{i\to j} = \mathrm{UFM}(\mathbf{I}_i, \mathbf{I}_j)$ from a pretrained flow model [32], and temporally aligned per-pixel 3D point maps $\{\mathbf{P}_t \in \mathbb{R}^{H\times W\times 3}\}$ from a pretrained reconstruction model [30]—across both position space and motion space. In a geometrically consistent video these two signals should agree after being expressed in a common

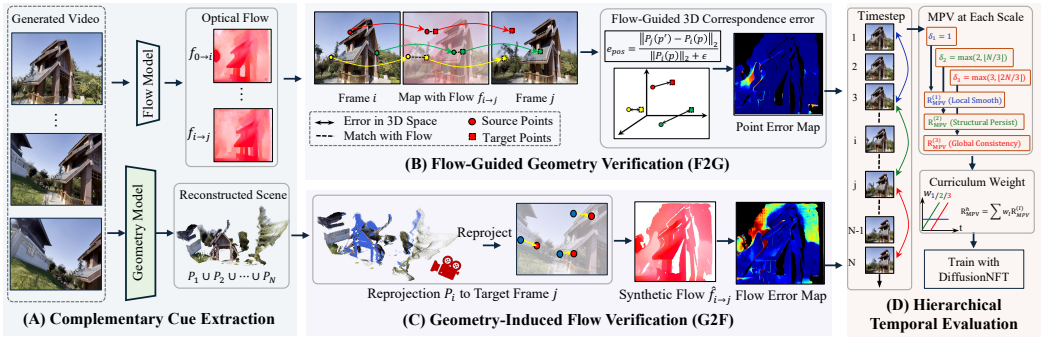

Figure 3: **Overview of WORLDPRISM**. Given a reference image, text prompt, and action sequence, the model autoregressively generates video clips using a flow-matching diffusion transformer. For each target clip, multiple candidates are sampled and evaluated by our proposed *Bidirectional Geometry-Flow Verification ((BGFV))* reward. BGFV performs bilateral cross-space verification : (i) a forward path lifts 2D optical flow into 3D to measure position-space consistency, and (ii) a reverse path projects reconstructed 3D geometry to synthesize flow for motion-space verification. The two signals are fused via a geometric mean and further aggregated through hierarchical temporal evaluation across multiple time offsets. The resulting reward guides reinforcement learning-based post-training to improve 3D geometric consistency without modifying the base model architecture.

coordinate frame; systematic disagreement constitutes evidence of genuine 3D violations rather than artifacts of any single perceptual model. The two verification paths are detailed below.

3.2.1 Flow-Guided Geometry Verification (F2G, Forward Path)

The forward path lifts 2D correspondences into 3D and evaluates their agreement in *position space*. This path directly addresses a key blind spot of unilateral 2D evaluation: geometric violations that are invisible after projection, such as depth ordering errors or lateral drift at similar screen positions, become explicit once correspondences are examined in full 3D coordinates.

Concretely, given a frame pair $(\mathbf{I}_i, \mathbf{I}_j)$ with estimated flow $\mathbf{f}_{i\to j}$, each pixel $\mathbf{p}$ in frame i is mapped to its corresponding location $\mathbf{p}' = \mathbf{p} + \mathbf{f}_{i\to j}(\mathbf{p})$ in frame j. We then look up the reconstructed 3D positions at both ends: $\mathbf{P}_i(\mathbf{p})$ from the source frame and $\mathbf{P}_j(\mathbf{p}')$ from the target frame, with both positions expressed in the same reconstructed coordinate frame. In a geometrically consistent video, these two 3D positions should coincide.

Since monocular reconstruction can produce point maps with an arbitrary per-pixel scale factor, direct Euclidean comparison is unreliable. We therefore adopt a scale-invariant relative distance:

$$e_{\text{pos}}(\mathbf{p}) = \frac{\|\mathbf{P}_j(\mathbf{p}') - \mathbf{P}_i(\mathbf{p})\|_2}{\|\mathbf{P}_i(\mathbf{p})\|_2 + \epsilon}. \tag{3}$$

This metric is inherently scale-invariant: for any positive scaling factor k, $\|k\mathbf{P}_j - k\mathbf{P}_i\|/\|k\mathbf{P}_i\| = \|\mathbf{P}_j - \mathbf{P}_i\|/\|\mathbf{P}_i\|$, naturally reducing the per-frame scale ambiguity of monocular reconstruction. Unlike depth-only metrics that measure only the z-component, this formulation captures lateral drift across the full XYZ space, providing a more complete constraint in position space. The path-level error averages over all valid correspondences: $E_{\text{pos}} = \frac{1}{|\Omega_{\text{pos}}|}\sum_{\mathbf{p}\in\Omega_{\text{pos}}} e_{\text{pos}}(\mathbf{p})$, where Ω_{pos} contains pixels with in-bound flow correspondences and valid reconstructed 3D points.

3.2.2 Geometry-Induced Flow Verification (G2F, Reverse Path)

The reverse path operates in the complementary direction: starting from 3D reconstruction, it synthesizes an expected 2D motion field and evaluates its agreement with the independently estimated optical flow in *motion space*. This path covers the blind spots of the forward path—if the reconstruction model produces locally consistent but globally incorrect geometry, the resulting synthetic flow will diverge from the independently observed motion, exposing the violation.

Concretely, given the reconstructed point maps $\mathbf{P}_i$, $\mathbf{P}_j$ and the relative camera transformation $[\mathbf{R} \mid \mathbf{t}]$ estimated between the two frames from the reconstructed geometry, we compute a geometric flow

field by projecting each 3D point in frame i into the coordinate system of frame j:

$$\hat{\mathbf{f}}_{i\to j}(\mathbf{p}) = \pi(\mathbf{R}\,\mathbf{P}_i(\mathbf{p}) + \mathbf{t}) - \mathbf{p}, \tag{4}$$

where $\pi(\cdot)$ denotes perspective projection. This geometric flow represents the 2D displacement that *would* be observed if the video depicted a geometrically consistent 3D scene undergoing the estimated camera motion. The verification error measures the pixel-level ℓ_2 discrepancy between this geometry-induced prediction and the independently estimated optical flow:

$$e_{\text{flow}}(\mathbf{p}) = \left\| \hat{\mathbf{f}}_{i\to j}(\mathbf{p}) - \mathbf{f}_{i\to j}(\mathbf{p}) \right\|_2. \tag{5}$$

Large residuals indicate that the 3D structure and the observed 2D motion are mutually inconsistent, exposing geometric violations such as incorrect depth ordering, object warping, or erroneous camera pose. The path-level error is averaged analogously: $E_{\text{flow}} = \frac{1}{|\Omega|}\sum_{\mathbf{p}\in\Omega} e_{\text{flow}}(\mathbf{p})$.

3.2.3 Bidirectional Fusion

The two paths are defined in different spaces with different error magnitudes: E_{pos} measures relative 3D displacement (typical range $\sim$0.05–0.5) while E_{flow} measures pixel-level residuals (typical range $\sim$2–20). We convert each to a reward via exponential mapping with per-path temperature parameters, and fuse the two via geometric mean to ensure that *both* paths must agree for a high reward:

$$R_{\text{bgfv}} = \sqrt{R_{\text{pos}} \cdot R_{\text{flow}}}, \quad R_{\text{pos}} = \exp(-\lambda_{\text{pos}} \cdot E_{\text{pos}}), \quad R_{\text{flow}} = \exp(-\lambda_{\text{flow}} \cdot E_{\text{flow}}), \tag{6}$$

The geometric mean is a deliberate choice: it yields a high reward only when both verification paths report low error, so a geometric violation that escapes detection in one space is still penalized if the complementary path flags it. This bidirectional fusion directly addresses the fundamental limitation of unilateral evaluation identified in §1. By spanning both position space and motion space with independently sourced signals, BGFV ensures that the blind spots of one projection direction are structurally covered by the other.

3.3 Hierarchical Temporal Evaluation

A single frame-pair offset cannot capture the full spectrum of 3D violations: adjacent frames reveal local jitter and deformation, while distant frames expose cumulative drift and global structural collapse. We therefore evaluate BGFV at three temporal scales, $\delta_1 = 1$, $\delta_2 = \max(2, \lfloor N/3 \rfloor)$, $\delta_3 = \max(3, \lfloor 2N/3 \rfloor)$, targeting local smoothness, structural persistence, and global consistency, respectively. At each level l, BGFV is averaged over all valid pairs with offset δ_l to yield $R^{(l)}_{\text{bgfv}}$. The levels are combined with curriculum-scheduled weights: $w_1 = 1, w_2 = \min(1,\ 2\rho), w_3 = \max(0,\ 2\rho - 1), R^{\text{hier}}_{\text{bgfv}} = \sum_{l=1}^{3} \bar{w}_l\, R^{(l)}_{\text{bgfv}}$, where $\rho = k/K$ is the training progress and $\bar{w}_l = w_l / \sum_{l'} w_{l'}$. This curriculum focuses early training on local consistency and progressively engages longer-range verification as the model stabilizes.

3.4 Dataset Curation

We curate a dedicated training set of 2k scenes from DL3DV [43], GTA-SFM [44], and MVS-Synth [45], collectively covering diverse indoor, outdoor, urban, and natural environments with accurate camera annotations. We employ Gemini 2.5 Pro [46] to generate high-quality text captions for each scene. During the training process, we adopt the complex camera poses from WorldCompass [35], which combines composite rotations and translations of varying difficulty to expose the model to challenging camera trajectories. More details are provided in Appendix D.1.

4 Experiments

4.1 Experimental Setup

Datasets. We build our evaluation benchmark on two datasets:(1) **DL3DV** [43], a real-world multi-view dataset featuring diverse indoor and outdoor scenes. Notably, the selected scenes are disjoint from those used during training, ensuring an unbiased evaluation; and (2) **MIND** [48],

Table 1: Quantitative comparison on the **DL3DV** dataset [43]. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
DiffusionAsShader [16]	0.0079	5.5012	2.4534	20.5061	0.4704	1.9719	0.6761
Trajectory-Attention [21]	0.0051	7.0090	1.0776	9.8631	0.1901	0.2217	0.7512
WorldPlay [3]	0.0022	2.8218	0.8715	10.2976	0.1585	2.4161	0.7193
w/ VideoGPA [12]	0.0025	1.9771	0.6178	4.8464	0.1311	3.7980	0.7594
w/ Epipolar error [13]	0.0019	2.0517	0.6283	4.5241	0.1993	3.4526	0.7218
w/ Depth reproj. [47]	0.0022	2.1463	0.5821	4.1173	0.1284	3.7215	0.7384
WorldCompass [35]	0.0049	2.8192	1.0067	18.2885	0.1782	3.2482	0.5938
WORLDPRISM (Ours)	0.0017	1.7158	0.5001	3.5592	0.1072	3.9554	0.7700

Table 2: Quantitative comparison on the **MIND** dataset [48]. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
DiffusionAsShader [16]	0.0071	5.3940	3.8087	90.0817	1.2890	0.5618	0.7466
Trajectory-Attention [21]	0.0049	6.6690	2.7520	38.0220	0.5027	0.2290	0.7999
WorldPlay [3]	0.0041	3.9612	2.0815	31.4461	0.5819	2.2986	0.7721
w/ VideoGPA [12]	0.0029	2.4350	0.9701	5.4022	0.3960	2.5700	0.8050
w/ Epipolar error [13]	0.0032	1.6799	0.9986	5.5603	0.5245	3.2527	0.8100
w/ Depth reproj. [47]	0.0037	1.5582	0.7975	4.6563	0.2791	3.6327	0.8255
WorldCompass [35]	0.0050	3.8168	1.5780	15.0367	0.4461	3.6289	0.8105
WORLDPRISM (Ours)	0.0021	1.5797	0.4626	3.0875	0.1058	3.5966	0.8513

a synthetic open-domain benchmark built with Unreal Engine 5, covering eight scene categories at 1080p resolution, designed to stress-test 3D consistency under complex geometry and frequent occlusions. We construct image-caption pairs from 300 scenes of DL3DV and 250 scenes of MIND, and design 8 types of complex camera trajectories as input poses for evaluation. Following §3.2, we utilize Gemini 2.5 Pro [46] to generate the captions. More details are provided in Appendix D.2.

Baselines and Metrics. We compare WORLDPRISM against two categories of baselines: (1) *Non-RL methods* that enforce geometric constraints at inference time, including DiffusionAsShader [16] and Trajectory-Attention [21]; and (2) *RL-based post-training approaches* built upon our base model WorldPlay [3], including VideoGPA [12] (a 2D perceptual evaluation reward), epipolar error [13] (a single geometric constraint), and depth reprojection error [47] (a unilateral depth-based reward). We additionally compare with WorldCompass [35], a concurrent RL-based method that focuses on 2D appearance consistency. For fair comparison, all RL-based variants adopt DiffusionNFT [33] as the optimization algorithm. We evaluate generated videos along two complementary axes: (1) *3D Consistency*, measured by reprojection error (Reproj.), epipolar error (Epipolar), bidirectional error (Bi err.), optical flow error (Flow), and point correspondence error (Point). (2) *Visual Quality*, assessed via human preference score (HPSv3) [49] and temporal structural similarity (T-SSIM).

Implementation Details. We build WORLDPRISM on top of WorldPlay, a flow-matching diffusion transformer operating in latent space. During post-training, we generate $G = 12$ candidate rollouts per training sample using the EMA policy and select the top-M and bottom-M ($M = 3$) samples for contrastive learning via DiffusionNFT [33]. The reward weights are set to $\alpha_{\text{bgfv}} = 1.0$, $\alpha_{\text{act}} = 1.0$, $\alpha_{\text{drift}} = 2.0$. The temperature parameters for bilateral fusion are $\lambda_{\text{pos}} = 5.0$ and $\lambda_{\text{flow}} = 0.5$. We use UFM [32] for optical flow estimation and Pi3 [30] for 3D point map reconstruction. Training is conducted on 16× GPUs for 2 days. The hierarchical temporal evaluation uses offsets $\delta_1 = 1$, $\delta_2 = \lfloor N/3 \rfloor$, $\delta_3 = \lfloor 2N/3 \rfloor$ with curriculum scheduling over the full training horizon.

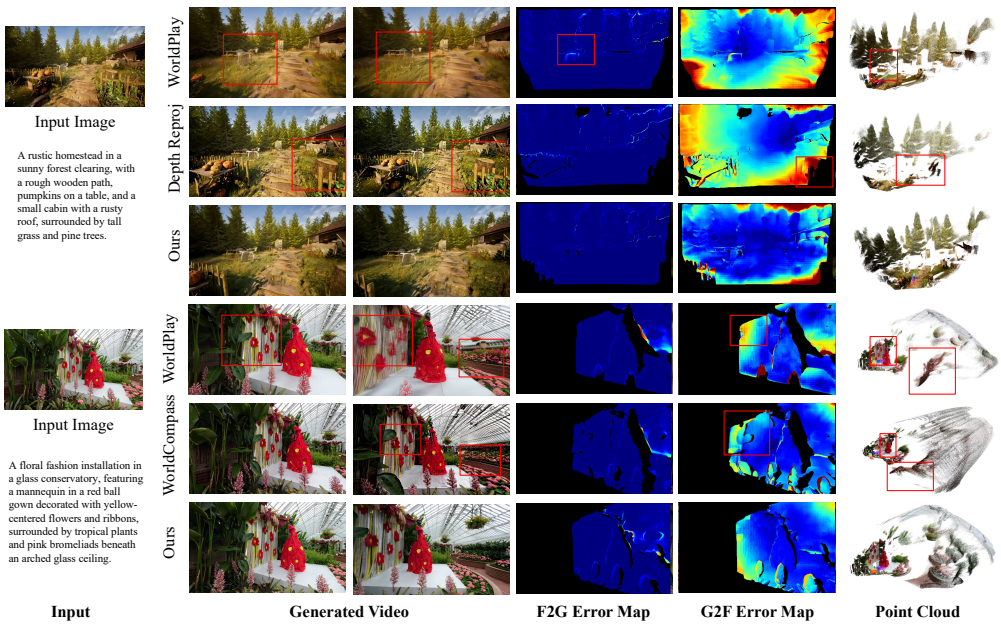

Figure 4: **Qualitative Comparison.** Key regions are highlighted with red bounding boxes. F2G Error Map measures the MSE of 3D point positions mapped via pseudo optical flow and reprojected back to the source frame. G2F Error Map measures the MSE error between the synthetic optical flow and the pseudo optical flow for each pixel on the source frame. Warmer colors indicate larger errors. **Note** that F2G errors are dominated by a few high-value outliers and most regions appear dark, whereas G2F errors have a narrower value range, yielding a brighter map.

4.2 Main Results

Quantitative Results. Tab. 1 and 2 report the main quantitative results on the DL3DV [43] and MIND [48] datasets. Equipped with our proposed Bilateral Cross-Space Verification, WORLDPRISM consistently outperforms a broad spectrum of baselines across almost all 3D consistency metrics, including (i) architecture-level approaches that inject geometric priors into the generator (DiffusionAsShader [16]), (ii) camera-conditioned methods relying on explicit trajectory control (Trajectory-Attention [21]), (iii) post-training methods explicitly targeting 3D consistency [12, 13, 47], and (iv) post-training methods optimizing only 2D appearance, such as WorldCompass [35].

Beyond the metrics directly aligned with our reward (Flow and Point), WORLDPRISM also delivers strong gains on metrics that are *not* part of the optimization objective, including Reproj., Epipolar, and Bi err. This indicates that bilateral verification captures a more general notion of 3D consistency, of which existing unilateral objectives can be viewed as special cases. Crucially, these geometric improvements do not come at the expense of perceptual quality: WORLDPRISM attains the best T-SSIM on both datasets and the best HPSv3 on DL3DV, while remaining competitive on MIND, confirming that the auxiliary reward terms effectively prevent quality regression during post-training.

Qualitative Results. Fig. 4 shows visual comparisons across representative scenes. In the first scene (rustic homestead), WorldPlay fails to preserve object permanence: the white object within the highlighted region disappears in subsequent frames, violating temporal and geometric consistency. In addition, the surrounding grass exhibits severe blurring, reflecting instability in fine-grained structures. For Depth Reprojection, the wooden stump in the highlighted region exhibits significant spatial drift in 3D. Although this misalignment is evident in the reconstructed geometry, it is less noticeable in 2D image space, revealing the limitation of single-view projection-based supervision. In contrast, our method preserves both object presence and spatial alignment by jointly enforcing consistency in both 2D and 3D spaces through complementary reward signals. In the second scene (glass conservatory), WorldPlay again produces noticeable blurring on the flowers attached to the wall, indicating a lack of structural stability. Depth Reprojection suffers from clear geometric distortion on the red brick wall behind the flower bed. WORLDPRISM mitigates these issues by enforcing cross-space consistency,

Table 3: Ablation study on the MIND dataset [48]. Each row removes one component from the full WORLDPRISM model. **Best** results are highlighted in darker blue, and **second best** in lighter blue.

Method	3D Consistency					Visual Quality	
	Reproj.↓	Epipolar↓	Bi err.↓	Flow↓	Point↓	HPSv3↑	T-SSIM↑
WORLDPRISM (Full)	0.0021	1.5797	0.4626	3.0875	0.1058	3.5966	0.8513
w/o F2G	0.0021	2.0509	0.7211	5.4574	0.1889	2.9173	0.8435
w/o G2F	0.0023	2.8806	0.8789	7.7379	0.1577	1.7200	0.7773
w/o Scale-Invariant	0.0023	2.2879	0.7531	6.7170	0.1327	2.5656	0.7973
w/o Hierarchical Temp.	0.0020	2.1544	0.8100	6.0527	0.2102	2.9556	0.8135

leading to sharper object boundaries and more coherent geometric layouts across frames. More visualization results are provided in Appendix E.

4.3 Ablation Study

To further validate the effectiveness of each core component, we conduct an ablation study on the MIND [48] dataset, as shown in Tab. 3.

(1) Flow-Guided Geometry Verification (Forward Path). Removing the forward path results in a clear degradation of 3D geometric accuracy, particularly in point-level accuracy and motion consistency. The point error increases from 0.1058 to 0.1889 (+78.5%), while the flow error rises from 3.0875 to 5.4574 (+76.7%). This substantial deterioration indicates that lifting 2D correspondences into 3D space is a critical component of our framework. It effectively constrains depth ordering and reduces lateral drift that cannot be reliably resolved in the 2D domain alone.

(2) Geometry-Induced Flow Verification (Reverse Path). Removing the reverse path also results in consistent performance degradation across multiple metrics. The epipolar error increases from 1.5797 to 2.8806, while the flow error increases from 3.0875 to 7.7379. In addition, visual quality HPSv3 decreases from 3.5966 to 1.7200. These results suggest that the reverse path plays a complementary role to the forward path by identifying globally inconsistent yet locally plausible 3D structures.

(3) Scale-Invariant Formulation. Replacing the scale-invariant formulation with direct Euclidean comparison degrades epipolar error from 1.58 to 2.29 and flow error from 3.09 to 6.72. This indicates that monocular scale ambiguity introduces systematic noise if not properly normalized.

(4) Hierarchical Temporal Evaluation. Using only adjacent-frame evaluation increases the point error (0.11→0.21) and increases bidirectional error (0.46→0.81). While single-offset evaluation captures local jitter, it fails to detect long-term drift, which is revealed by hierarchical multi-scale offsets.

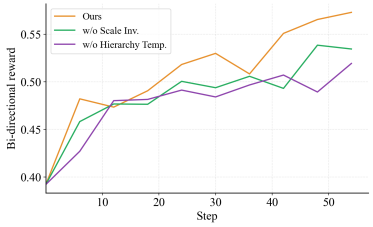

Figure 5: **Reward score curves during training.**

To further prove the effectiveness of our bidirectional design, we visualize the 3D consistency error maps under both two paths. As shown in Fig. 2, either single path fails to detect the 3D inconsistencies: (1) The flow-based path (G2F) fails to detect structural distortions; (2) The geometry-verified path (F2G) fails to detect visual artifacts. In contrast, our reward leverages complementary signals from both paths to successfully detect 3D inconsistencies in the generated video. The reward curve during training, shown in Fig. 5, further demonstrates the effectiveness of our proposed Scale-Invariant Formulation and Hierarchical Temporal Evaluation.

5 Conclusion

We presented WORLDPRISM, a post-training framework that improves the 3D consistency of video world models via Bidirectional Geometry-Flow Verification, which cross-checks optical flow and 3D reconstruction through two complementary paths (F2G and G2F) so that the blind spots of each single evaluation space are structurally covered by the other. Combined with scale-invariant metrics and hierarchical temporal evaluation, our reward provides a robust 3D consistency signal that avoids the

systematic limitations of unilateral 2D-based approaches. We build our WORLDPRISM with WorldPlay as the base model, and experiments on diverse benchmarks show that WORLDPRISM achieves state-of-the-art 3D consistency while preserving visual quality and generative diversity. We present bidirectional cross-space verification as a preliminary step toward more robust reward design in video generation, and we hope that the proposed WORLDPRISM serves as a useful reference for future efforts in building 3D-consistent video world models.

Acknowledgments and Disclosure of Funding

The authors did not receive any third-party funding or support for this work. The authors declare no competing interests.

References

[1] Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.

[2] Sherry Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators, 2024.

[3] Wenqiang Sun, Haiyu Zhang, Haoyuan Wang, Junta Wu, Zehan Wang, Zhenwei Wang, Yunhong Wang, Jun Zhang, Tengfei Wang, and Chunchao Guo. Worldplay: Towards long-term geometric consistency for real-time interactive world modeling. *arXiv preprint arXiv:2512.14614*, 2025.

[4] Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. Hunyuanvideo: A systematic framework for large video generative models, 2025.

[5] Team Wan. Wan: Open and advanced large-scale video generative models, 2025.

[6] NVIDIA. Cosmos world foundation model platform for physical ai, 2025.

[7] Jiahao Lu, Minghao Yin, Wenbo Hu, Hengyu Liu, Wang Zhao, Sai-Kit Yeung, Ying Shan, and Yuan Liu. Gae: Learning a geometry-native latent space for 3d-consistent world generation. *arXiv preprint arXiv:2609.24981*, 2026.

[8] Haoyi Duan, Hong-Xing Yu, Sirui Chen, Li Fei-Fei, and Jiajun Wu. Worldscore: A unified evaluation benchmark for world generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 27713–27724, 2025.

[9] Mohammad Asim, Christopher Wewer, Thomas Wimmer, Bernt Schiele, and Jan Eric Lenssen. MEt3R: Measuring multi-view consistency in generated images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[10] Yiran Qin, Zhelun Shi, Jiwen Yu, Xijun Wang, Enshen Zhou, Lijun Li, Zhenfei Yin, Xihui Liu, Lu Sheng, Jing Shao, Lei Bai, Wanli Ouyang, and Ruimao Zhang. Worldsimbench: Towards video generation models as world simulators. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025.

[11] Yunpeng Bai, Shaoheng Fang, Chaohui Yu, Fan Wang, and Qixing Huang. Geovideo: Introducing geometric regularization into video generation model. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

[12] Hongyang Du, Junjie Ye, Xiaoyan Cong, Runhao Li, Jingcheng Ni, Aman Agarwal, Zeqi Zhou, Zekun Li, Randall Balestriero, and Yue Wang. Videogpa: Distilling geometry priors for 3d-consistent video generation. *arXiv preprint arXiv:2601.23286*, 2026.

[13] Orest Kupyn, Fabian Manhardt, Federico Tombari, and Christian Rupprecht. Epipolar geometry improves video generation models. *arXiv preprint arXiv:2510.21615*, 2025.

[14] Minghao Yin, Jiahao Lu, Wenbo Hu, Wang Zhao, Ying Shan, and Kai Han. Scope: sightline-coordinate positional encoding for video diffusion transformers. *arXiv preprint arXiv:2606.27345*, 2026.

[15] Qihang Zhang, Shuangfei Zhai, Miguel Angel Bautista, Kevin Miao, Alexander Toshev, Joshua Susskind, and Jiatao Gu. World-consistent video diffusion with explicit 3d modeling. *arXiv preprint arXiv:2412.01821*, 2024.

[16] Zekai Gu, Rui Yan, Jiahao Lu, Peng Li, Zhiyang Dou, Chenyang Si, Zhen Dong, Qifeng Liu, Cheng Lin, Ziwei Liu, et al. Diffusion as shader: 3d-aware video diffusion for versatile video generation control. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, pages 1–12, 2025.

[17] Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025.

[18] Zeqi Xiao, Yushi Lan, Yifan Zhou, Wenqi Ouyang, Shuai Yang, Yanhong Zeng, and Xingang Pan. Worldmem: Long-term consistent world simulation with memory, 2026.

[19] Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. Cameractrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*, 2024.

[20] Hao He, Ceyuan Yang, Shanchuan Lin, Yinghao Xu, Meng Wei, Liangke Gui, Qi Zhao, Gordon Wetzstein, Lu Jiang, and Hongsheng Li. Cameractrl ii: Dynamic scene exploration via camera-controlled video diffusion models. *arXiv preprint arXiv:2503.10592*, 2025.

[21] Zeqi Xiao, Wenqi Ouyang, Yifan Zhou, Shuai Yang, Lei Yang, Jianlou Si, and Xingang Pan. Trajectory attention for fine-grained video motion control. *arXiv preprint arXiv:2411.19324*, 2024.

[22] Weijie Wang, Xiaoxuan He, Youping Gu, Zeyu Zhang, Yefei He, Yanbo Ding, Xirui Hu, Donny Y. Chen, Zhiyuan He, Yuqing Yang, Yifan Yang, and Bohan Zhuang. World-r1: Reinforcing 3d constraints for text-to-video generation. *arXiv preprint arXiv:2604.24764*, 2026.

[23] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2023.

[24] Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

[25] Jie Liu, Gongye Liu, Jiajun Liang, Yangguang Li, Jiaheng Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Wanli Ouyang. Flow-grpo: Training flow matching models via online rl, 2025.

[26] Zeyue Xue, Jie Wu, Yu Gao, Fangyuan Kong, Lingting Zhu, Mengzhao Chen, Zhiheng Liu, Wei Liu, Qiushan Guo, Weilin Huang, and Ping Luo. Dancegrpo: Unleashing grpo on visual generation, 2025.

[27] Bram Wallace, Meihua Dang, Rafael Rafailov, Linqi Zhou, Aaron Lou, Senthil Purushwalkam, Stefano Ermon, Caiming Xiong, Shafiq Joty, and Nikhil Naik. Diffusion model alignment using direct preference optimization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[28] Paul D Sampson. Fitting conic sections to "very scattered" data: An iterative refinement of the bookstein algorithm. *Computer graphics and image processing*, 18(1):97–108, 1982.

[29] Ayush Sarkar, Hanlin Mai, Amitabh Mahapatra, Svetlana Lazebnik, David A Forsyth, and Anand Bhattad. Shadows don't lie and lines can't bend! generative models don't know projective geometry... for now. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28140–28149, 2024.

[30] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. π^3: Permutation-equivariant visual geometry learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.

[31] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views. In *International Conference on Learning Representations (ICLR)*, 2026.

[32] Yuchen Zhang, Nikhil Keetha, Chenwei Lyu, Bhuvan Jhamb, Yutian Chen, Yuheng Qiu, Jay Karhade, Shreyas Jha, Yaoyu Hu, Deva Ramanan, Sebastian Scherer, and Wenshan Wang. UFM: A simple path towards unified dense correspondence with flow. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[33] Kaiwen Zheng, Huayu Chen, Haotian Ye, Haoxiang Wang, Qinsheng Zhang, Kai Jiang, Hang Su, Stefano Ermon, Jun Zhu, and Ming-Yu Liu. Diffusionnft: Online diffusion reinforcement with forward process. *ICLR*, 2026.

[34] Bing Wu, Chang Zou, Changlin Li, Duojun Huang, Fang Yang, Hao Tan, Jack Peng, Jianbing Wu, Jiangfeng Xiong, Jie Jiang, et al. Hunyuanvideo 1.5 technical report. *arXiv preprint arXiv:2511.18870*, 2025.

[35] Zehan Wang, Tengfei Wang, Haiyu Zhang, Xuhui Zuo, Junta Wu, Haoyuan Wang, Wenqiang Sun, Zhenwei Wang, Chenjie Cao, Hengshuang Zhao, Chunchao Guo, and Zhou Zhao. Worldcompass: Reinforcement learning for long-horizon world models. *arXiv preprint arXiv:2602.09022*, 2026.

[36] Zachary Teed and Jia Deng. DROID-SLAM: Deep visual slam for monocular, stereo, and RGB-D cameras. *Advances in Neural Information Processing Systems*, 34:16558–16569, 2021.

[37] Leslie Gu, Junhwa Hur, Charles Herrmann, Fangneng Zhan, Todd Zickler, Deqing Sun, and Hanspeter Pfister. Geco: A differentiable geometric consistency metric for video generation. *arXiv preprint*, 2025.

[38] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

[39] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[40] Tengjiao Yin, Jinglei Shi, Heng Guo, and Xi Wang. Vigor: Video geometry-oriented reward for temporal generative alignment. *arXiv preprint arXiv:2603.16271*, 2025.

[41] Jiahao Lu, Tianyu Huang, Peng Li, Zhiyang Dou, Cheng Lin, Zhiming Cui, Zhen Dong, Sai-Kit Yeung, Wenping Wang, and Yuan Liu. Align3r: Aligned monocular depth estimation for dynamic videos. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22820–22830. IEEE, 2025.

[42] Jiahao Lu, Jiayi Xu, Wenbo Hu, Ruijie Zhu, Chengfeng Zhao, Sai-Kit Yeung, Ying Shan, and Yuan Liu. Track4world: Feedforward world-centric dense 3d tracking of all pixels. In *European Conference on Computer Vision*, pages 184–204. Springer, 2026.

[43] Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yanqi Lu, et al. DL3DV-10K: A large-scale scene dataset for deep learning-based 3d vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[44] Kaixuan Wang and Shaojie Shen. Flow-motion and depth network for monocular stereo and beyond. In *IEEE Robotics and Automation Letters*, volume 5, pages 3307–3314, 2020.

[45] Po-Han Huang, Kevin Matzen, Johannes Kopf, Narendra Ahuja, and Jia-Bin Huang. DeepMVS: Learning multi-view stereopsis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2821–2830, 2018.

[46] Gemini Team, Rohan Anil, Sebastian Borgeaud, et al. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2024.

[47] Zhaochong An, Orest Kupyn, Théo Uscidda, Andrea Colaco, Karan Ahuja, Serge Belongie, Mar Gonzalez-Franco, and Marta Tintore Gazulla. Vggrpo: Towards world-consistent video generation with 4d latent reward. *arXiv preprint arXiv:2603.26599*, 2026.

[48] Yixuan Ye, Xuanyu Lu, Yuxin Jiang, Yuchao Gu, Rui Zhao, Qiwei Liang, Jiachun Pan, Fengda Zhang, Weijia Wu, and Alex Jinpeng Wang. Mind: Benchmarking memory consistency and action control in world models. *arXiv preprint arXiv:2602.08025*, 2026.

[49] Yuhang Ma, Xiaoshi Wu, Keqiang Sun, and Hongsheng Li. Hpsv3: Towards wide-spectrum human preference score. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15086–15095, 2025.

[50] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. VBench: Comprehensive benchmark suite for video generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[51] David G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2):91–110, 2004.

[52] Jürgen Sturm, Nikolas Engelhard, Felix Endres, Wolfram Burgard, and Daniel Cremers. A benchmark for the evaluation of RGB-D SLAM systems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 573–580, 2012.

[53] Haofei Xu, Jing Zhang, Jianfei Cai, Hamid Rezatofighi, Fisher Yu, Dacheng Tao, and Andreas Geiger. Unifying flow, stereo and depth estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(11):13941–13958, 2023.

[54] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs field transforms for optical flow. In *European Conference on Computer Vision (ECCV)*, pages 402–419, 2020.

[55] Daniel J. Butler, Jonas Wulff, Garrett B. Stanley, and Michael J. Black. A naturalistic open source movie for optical flow evaluation. In *European Conference on Computer Vision (ECCV)*, pages 611–625, 2012.

[56] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[57] Yaofang Liu, Xiaodong Cun, Xuebo Liu, Xintao Wang, Yong Zhang, Haoxin Chen, Yang Liu, Tieyong Zeng, Raymond Chan, and Ying Shan. EvalCrafter: Benchmarking and evaluating large video generation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22139–22149, 2024.

A Additional Quantitative Results

To further validate that our post-training procedure does not compromise general video-generation capabilities, we evaluate WorldPrism and baselines on VBench [50], a comprehensive benchmark suite that covers diverse aspects of video quality beyond 3D consistency. As shown in Tab. 4, WorldPrism achieves the best or second-best performance across nearly all dimensions. Compared to our base model WorldPlay [3], WorldPrism yields substantial improvements on consistency-related metrics (Subject Consistency: $0.9104 \rightarrow 0.9483$, Background Consistency: $0.9117 \rightarrow 0.9255$) while simultaneously improving visual quality indicators such as Aesthetic Quality ($0.4791 \rightarrow 0.4926$) and I2V fidelity. This confirms that the auxiliary reward terms (R_{act} and R_{drift}) effectively prevent quality degradation during geometric-focused post-training. Compared to WorldCompass [35], WorldPrism delivers notably stronger Subject Consistency (0.9483 vs 0.8969) and Background Consistency (0.9255 vs 0.8991), showing that enforcing explicit 3D geometric constraints translates to more coherent temporal consistency as perceived by VBench's evaluation protocols. These results indicate that our Bidirectional Geometry-Flow Verification steers the model toward genuine 3D-consistent generation rather than superficial appearance matching, and that such geometric grounding benefits overall video quality rather than trading it off.

Table 4: Evaluation on public benchmark VBench [50].

Method	I2V Subject	I2V Background	Subject Consistency	Background Consistency	Motion Smooth	Aesthetic Quality	Imaging Quality
DiffusionAsShader [16]	0.9443	0.9618	0.8957	0.9159	0.9828	0.4764	0.6250
Trajectory-Attention [21]	0.9481	0.9622	0.9314	0.9324	0.9859	0.4699	0.6167
WorldPlay [3]	0.9736	0.9767	0.9104	0.9117	0.9916	0.4791	0.6780
WorldCompass [35]	0.9732	0.9745	0.8969	0.8991	0.9884	0.4908	0.7164
WORLDPRISM (Ours)	0.9822	0.9820	0.9483	0.9255	0.9923	0.4926	0.6896

B Extended Evaluations

B.1 Independent Evaluation of Geometric Consistency

Several 3D-consistency metrics in the main paper are only partially independent of the reward models: Reproj. relies on geometry predicted by Pi3 [30], while Flow, Point, and Bi err. rely on Pi3 and UFM [32], which are also used to construct the training reward (Epipolar uses SIFT [51] correspondences and is independent). To verify that the improvements are not specific to the Pi3/UFM reward space, we conduct two additional evaluations.

Evaluation against prescribed camera trajectories. Generated videos have no pixel-aligned ground-truth point maps or optical flow, but the prescribed camera trajectories injected as conditioning signals provide direct ground-truth camera poses. We estimate camera poses from each generated video with the held-out VGGT [39] and compute the Absolute Trajectory Error (ATE) and Relative Rotation Error (RRE) [52] against the prescribed trajectory, which directly measures whether the generated camera motion follows the input actions.

Evaluation with held-out perception models. We further evaluate geometric consistency with perception models that are not involved in reward construction: UniMatch [53] and RAFT [54] for correspondence estimation, and VGGT [39] and Depth Anything 3 (DA3) [31] for geometry estimation. *UniMatch / RAFT Sampson* report the mean epipolar (Sampson) error [28] of the estimated correspondences under the ground-truth camera poses. *VGGT / DA3 Depth* measure cross-frame depth inconsistency after warping the predicted depth with the ground-truth camera motion. *VGGT+UniMatch / VGGT+RAFT EPE* measure the endpoint error between the VGGT-induced rigid flow and the flow estimated by UniMatch or RAFT.

As shown in Tab. 5, WORLDPRISM achieves the best results on all metrics. Relative to the second-best baseline, it reduces ATE and RRE by 2.4% and 5.1%, Sampson error by up to 6.0%, depth inconsistency by up to 43.9%, and geometry-flow EPE by up to 6.5%. Measured as the average relative improvement over DiffusionAsShader [16], WORLDPRISM delivers a gain of 51.64%, outperforming WorldPlay [3] (38.10%) and WorldCompass [35] (44.76%). These results confirm that the geometric improvements transfer to independent geometry estimators and to ground-truth camera trajectories, rather than reflecting overfitting to the reward models.

B.2 User Study

We conduct a user study with 22 participants, who evaluate the generated videos along four criteria: geometric consistency, temporal stability, camera-motion alignment, and visual quality. For each scene, four anonymized

Table 5: Independent evaluation of geometric consistency against prescribed camera trajectories and with held-out perception models (VGGT [39], Depth Anything 3 [31], UniMatch [53], RAFT [54]) that are not used for reward construction. Avg. Gain is the average relative improvement over DiffusionAsShader across all metrics.

Method	Camera Pose		Sampson Error		Depth Inconsistency		Geometry-Flow EPE		Avg.
	ATE ↓	RRE ↓	UniMatch ↓	RAFT ↓	VGGT ↓	DA3 ↓	VGGT+UniMatch ↓	VGGT+RAFT ↓	Gain ↑
DiffusionAsShader [16]	0.0421	17.4198	29.710	28.880	0.4209	0.37475	97.359	95.495	–
WorldPlay [3]	0.0590	6.9930	15.877	15.972	0.2493	0.21594	43.754	42.378	+38.10%
WorldCompass [35]	0.0448	5.8634	15.901	15.787	0.2510	0.21590	37.558	36.141	+44.76%
WORLDPRISM (Ours)	0.0411	5.5628	14.924	15.273	0.1398	0.18061	35.127	35.094	+51.64%

videos (one per method) are presented in random order, and participants select the best two under one assigned criterion. The score of each method is computed as $5\times$ its selection rate, and Overall aggregates all four criteria.

As shown in Tab. 6, WORLDPRISM achieves the highest scores in geometric consistency, temporal stability, visual quality, and overall preference, while remaining competitive in camera-motion alignment. Since participants are given only a coarse motion type rather than the reference trajectory, camera-motion alignment reflects perceived motion plausibility rather than precise trajectory-following accuracy. This explains why WorldPlay may score higher on this criterion despite its larger ATE and RRE in Tab. 5. Overall, the results show that WORLDPRISM substantially improves the perceived 3D coherence and subjective quality of the generated videos.

Table 6: User study with 22 participants. Scores are computed as $5\times$ the selection rate; Overall aggregates all four criteria.

Method	Geometric Consistency	Temporal Stability	Camera-Motion Alignment	Visual Quality	Overall
DiffusionAsShader [16]	0.99	0.88	1.14	0.81	0.95
WorldPlay [3]	2.89	3.44	3.42	3.06	3.16
WorldCompass [35]	2.30	2.06	2.63	2.50	2.41
WORLDPRISM (Ours)	3.82	3.63	2.81	3.63	3.48

B.3 Occlusion Handling

WORLDPRISM handles occlusion through covisibility filtering and geometric-validity checks in both F2G and G2F. For each source–target frame pair, UFM [32] predicts optical flow together with a source-frame covisibility mask M_{cov}, which indicates whether each source pixel has a reliable correspondence in the target frame. UFM's covisibility head is pretrained on large-scale correspondence data and is designed to identify occluded, out-of-view, and otherwise unreliable matches. Both branches retain only pixels satisfying

$$M_{\text{cov}}(\mathbf{p}) > \tau_{\text{cov}}, \tag{7}$$

where τ_{cov} is a fixed threshold, and all errors are averaged only over the retained pixels.

In F2G, the 3D correspondence error is computed only when the flow-mapped target location lies inside the image and both the source and target 3D points are valid. In G2F, the flow error is computed only when the projected 3D point has positive depth, lies inside the target image, and produces a finite residual. Therefore, pixels identified as non-covisible or occluded by M_{cov}, together with out-of-view or geometrically invalid pixels detected by these validity checks, are excluded from the error computation. This prevents valid occlusions from being incorrectly penalized.

To evaluate the reliability of this filtering mechanism, we compare UFM's predicted covisibility masks with ground-truth visibility masks derived from the MPI Sintel [55] occlusion annotations. As shown in Tab. 7, across 100 valid frame pairs covering all 50 scenes, the predicted masks show strong agreement with ground-truth visibility (96.92% Dice and 99.20% recall), indicating that the pretrained covisibility head reliably identifies pixels with valid cross-frame correspondences and substantially reduces erroneous penalties caused by occlusion.

Table 7: Agreement between UFM's predicted covisibility masks and ground-truth visibility masks derived from MPI Sintel [55] occlusion annotations (100 frame pairs covering all 50 scenes).

Covisible Dice ↑	Covisible IoU ↑	Precision ↑	Recall ↑	Pixel Accuracy ↑
96.92%	94.03%	94.75%	99.20%	94.13%

B.4 Evaluation on Dynamic Scenes

The static-scene assumption is adopted to isolate geometry-induced inconsistencies during training, rather than implying that the post-training framework is fundamentally restricted to static content. Although WORLDPRISM does not explicitly disentangle ego-motion from object motion, the reward primarily enforces the geometric stability of the static environment. To examine its behavior on dynamic content, we evaluate WORLDPRISM and WorldPlay on dynamic scenes from VBench [50] under both static- and moving-camera settings, reporting standard consistency, motion, and video quality metrics [50, 56, 57].

As shown in Tab. 8, WORLDPRISM improves over WorldPlay on all metrics in both settings. In particular, under moving-camera generation, WORLDPRISM achieves higher Dynamic Degree ($0.9625 \rightarrow 0.9875$) and Motion Smoothness ($0.9714 \rightarrow 0.9746$), indicating that the geometric reward does not suppress object dynamics. It also improves Temporal SSIM ($0.6658 \rightarrow 0.6827$), indicating better temporal stability of the generated videos. Explicitly decoupling dynamic foreground from static background remains a promising direction (see Sec. F).

Table 8: Evaluation on dynamic scenes from VBench [50] under static- and moving-camera settings. Top: VBench consistency, motion, and quality metrics. Bottom: low-level temporal metrics [56, 57]. Better results within each camera setting are in **bold**.

Model	Subject Consistency ↑	Background Consistency ↑	Dynamic Degree ↑	Motion Smoothness ↑	Aesthetic Quality ↑	Imaging Quality ↑
Static camera						
WorldPlay [3]	0.9674	0.9658	0.9653	0.9934	0.6165	0.7088
WORLDPRISM (Ours)	**0.9878**	**0.9757**	**0.9785**	**0.9957**	**0.6285**	**0.7103**
Moving camera						
WorldPlay [3]	0.8552	0.8906	0.9625	0.9714	0.5317	0.6463
WORLDPRISM (Ours)	**0.9294**	**0.9284**	**0.9875**	**0.9746**	**0.5847**	**0.7219**

Model	Flicker ↓	Temporal SSIM ↑	Warp Error ↓	Flow Magnitude ↑	Sharpness ↑
Static camera					
WorldPlay [3]	0.0067	0.9683	0.0067	0.5200	0.0190
WORLDPRISM (Ours)	**0.0041**	**0.9838**	**0.0041**	**0.6000**	**0.0325**
Moving camera					
WorldPlay [3]	0.0520	0.6658	0.0520	5.1700	0.0126
WORLDPRISM (Ours)	**0.0514**	**0.6827**	**0.0514**	**7.9500**	**0.0352**

C Details of the WORLDPRISM

C.1 Post-Training via DiffusionNFT

Reward composition. Optimizing solely for geometric consistency risks degrading action fidelity or visual quality, as the model may exploit degenerate solutions (static frames or minimal camera motion) that trivially minimize geometric error. We therefore combine the hierarchical BGFV signal with two auxiliary objectives adopted from WorldCompass [35]:

$$R = \alpha_{\text{bgfv}}\, R^{\text{hier}}_{\text{bgfv}} + \alpha_{\text{act}}\, R_{\text{act}} + \alpha_{\text{drift}}\, R_{\text{drift}}, \tag{8}$$

where R_{act} measures action-following accuracy by comparing the camera motion estimated from the generated clip against the conditioning action sequence, ensuring that the model faithfully executes the instructed camera trajectory rather than defaulting to geometrically trivial motions. R_{drift} penalizes visual quality degradation relative to the conditioning frame by measuring perceptual similarity, preventing the model from sacrificing texture fidelity, sharpness, or color consistency in pursuit of geometric accuracy. Together, these auxiliary signals form a multi-objective optimization landscape that prevents the model from exploiting geometric consistency at the expense of action fidelity or perceptual quality, maintaining a balanced improvement across all axes.

Optimization. We adopt DiffusionNFT [33], a negative-aware fine-tuning algorithm specifically designed for flow matching models. Unlike standard RLHF approaches that reinforce only high-reward samples, DiffusionNFT leverages both high- and low-reward samples through a contrastive learning framework, enabling more sample-efficient optimization by simultaneously pulling the model toward desirable outputs and pushing it away from undesirable ones.

Given G rollout clips $\{x_n^{(i)}\}_{i=1}^G$ generated by the EMA policy $\pi_{\theta_{\text{old}}}$ for a given training sample, each clip receives per-channel reward scores $\{s_j^{(i)}\}_{j\in\{\text{bgfv,act,drift}\}}$. We first compute per-channel advantages via z-score normalization to handle the different scales of reward signals:

$$a_j^{(i)} = \frac{s_j^{(i)} - \text{mean}(\{s_j^{(i)}\}_{i=1}^G)}{\text{std}(\{s_j^{(i)}\}_{i=1}^G)}, \tag{9}$$

which centers each reward channel to zero mean and unit variance within each batch of rollouts, removing the need for manual reward scaling. The normalized advantages are then aggregated into a single optimality weight:

$$r^{(i)} = \frac{1}{2} + \frac{1}{2}\,\text{clip}\left[\frac{\sum_j \alpha_j\, a_j^{(i)}}{Z},\ -1,\ 1\right], \tag{10}$$

where $Z = \sum_j \alpha_j$ is a normalizing constant. The resulting $r^{(i)} \in [0, 1]$ serves as a soft interpolation coefficient: values near 1 indicate high-quality samples that should be reinforced, while values near 0 indicate low-quality samples that should be suppressed.

From the G candidates, we select the top-M and bottom-M by total reward to form a contrastive training subset $\mathcal{G}_{\text{sub}}$, discarding intermediate samples that provide a weaker learning signal. For each selected sample, we construct noisy latents at randomly subsampled timesteps $t \sim \mathcal{T}_{\text{sub}}$:

$$\mathbf{z}_t^{(i)} = (1-t)\, x_n^{(i)} + t\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{11}$$

and define implicit positive/negative velocity targets through an interpolation between the EMA policy and the current model:

$$v_\theta^+ = (1-\beta)\, v_{\theta_{\text{old}}}(\mathbf{z}_t^{(i)}, t) + \beta\, v_\theta(\mathbf{z}_t^{(i)}, t), \tag{12}$$

$$v_\theta^- = (1+\beta)\, v_{\theta_{\text{old}}}(\mathbf{z}_t^{(i)}, t) - \beta\, v_\theta(\mathbf{z}_t^{(i)}, t), \tag{13}$$

where $v_{\theta_{\text{old}}}$ and v_θ denote predictions from the EMA policy and the current model respectively, and β controls the interpolation strength. Intuitively, v_θ^+ represents a target that slightly amplifies the current model's deviation from the EMA baseline (reinforcing its predictions), while v_θ^- represents a mirror target that penalizes the current deviation (suppressing its predictions).

The final training objective combines both positive and negative supervision weighted by the optimality coefficient:

$$\mathcal{L}(\theta) = \mathbb{E}_{\substack{i\sim\mathcal{G}_{\text{sub}} \\ t\sim\mathcal{T}_{\text{sub}}}}\left[r^{(i)}\left\|v_\theta^+ - v^{(i)}\right\|_2^2 + (1-r^{(i)})\left\|v_\theta^- - v^{(i)}\right\|_2^2\right], \tag{14}$$

where $v^{(i)} = x_n^{(i)} - \boldsymbol{\epsilon}$ is the ground-truth velocity target derived from the clean sample and the noise. For high-reward samples ($r^{(i)} \approx 1$), the loss is dominated by the positive term, reinforcing the current model's velocity predictions. For low-reward samples ($r^{(i)} \approx 0$), the negative term dominates, optimizing toward the mirror velocity v_θ^- and effectively steering the model away from generating low-quality outputs. This asymmetric contrastive formulation enables the model to learn from both success and failure modes simultaneously, leading to more stable and efficient convergence than purely positive-reinforcement approaches.

D Experimental Settings

D.1 Details of Training Dataset Curation

Our training set is curated from three complementary sources that collectively provide diverse real-world and synthetic scenes, totaling 2,003 video sequences.

DL3DV [43]. DL3DV-10K is a large-scale real-world scene dataset containing over 10,000 videos captured at 4K resolution, covering 65 diverse categories including both indoor and outdoor environments (e.g., offices, streets, museums, parks). We select 1,000 representative scenes from this dataset (49.9% of our training set), prioritizing diversity in scene layout, scale, and appearance. These real-world sequences provide naturalistic camera trajectories and complex visual phenomena such as reflections, transparency, and repetitive textures.

GTA-SFM [44]. GTA-SFM is a synthetic dataset rendered from the Grand Theft Auto V (GTA V) game engine, featuring a variety of urban driving scenarios with diverse environmental conditions, including different weather, time of day, and lighting. We utilize 643 sequences from this dataset (32.1%), which provide geometrically consistent multi-view imagery under controlled settings that complement the real-world data.

MVS-Synth [45]. MVS-Synth is another synthetic dataset rendered from the GTA V engine, specifically designed for multi-view stereo research. It contains 120 urban scenes, each comprising 100 high-resolution RGB frames (1920×1080), covering challenging visual conditions such as textureless surfaces, reflective materials, and thin structures. We include 360 sequences from this dataset (18.0%), leveraging its diverse urban environments to improve generalization in scenarios with limited real-world data coverage.

D.2 Details of Benchmark Setup

Our evaluation benchmark comprises 550 scenes from two sources: 300 scenes from DL3DV [43] (disjoint from the training split) and 250 scenes from MIND [48]. For each scene, we design 16 types of camera trajectories, ranging from simple single-axis motions to complex compound motions, as summarized in Table 9.

Table 9: The 16 camera trajectory types used in our evaluation benchmark.

ID	Trajectory	ID	Trajectory
0	Move Forward	8	Forward + Turn Left
1	Move Left	9	Forward + Look Up
2	Turn Left	10	Orbit Left
3	Look Up	11	Zigzag
4	Forward-Backward	12	U-Turn
5	Left-Right	13	Pan 360
6	Turn Left-Right	14	Crane Up
7	Look Up-Down	15	Spiral

These trajectories range from single-axis translations/rotations (ID 0–3) to round-trip motions (ID 4–7), combined motions (ID 8–9), and complex multi-axis trajectories (ID 10–15), enabling fine-grained diagnosis of 3D consistency as motion complexity increases.

E Additional Qualitative Results

We provide additional qualitative results in Fig. 6–9 to further demonstrate the effectiveness of our proposed WORLDPRISM in improving the 3D consistency of videos generated by the HunyuanVideo model [4] through post-training. For each comparison, we visualize the generated video frames, F2G error maps, G2F error maps, and reconstructed 3D point clouds.

As shown in Fig. 6 and Fig. 7, the baseline methods (WorldPlay [3], VideoGPA [12], and Depth Reprojection from VGGRPO [47]) exhibit noticeably warmer colors in both the F2G and G2F error maps, indicating larger geometric inconsistencies in the generated videos. In contrast, WORLDPRISM produces substantially lower error across both verification paths, confirming that the bilateral cross-space reward effectively enforces mutual agreement between 2D motion and 3D structure. The reconstructed point clouds further corroborate these observations: baselines produce fragmented or distorted geometry, whereas WORLDPRISM yields coherent and structurally plausible 3D reconstructions.

Fig. 8 and Fig. 9 present direct frame-level comparisons across diverse scenes. Compared with WorldPlay [3], VideoGPA [12], and Depth Reprojection from VGGRPO [47], videos generated by WORLDPRISM exhibit fewer structural distortions, reduced spatial drift, and better-preserved object boundaries under complex camera motions. These improvements are consistent across both indoor and outdoor scenarios, demonstrating that the bidirectional design of BGFV generalizes well to a wide variety of scene types and camera trajectories without sacrificing visual quality.

F Limitation

Our method is primarily designed for static scenes where all observed optical flow is induced by camera motion. When the generated video contains dynamic objects (e.g., moving pedestrians or vehicles), the current framework cannot distinguish whether the observed flow originates from camera ego-motion or independent object motion. This ambiguity leads to inaccurate reward signals: the Bidirectional Geometry-Flow Verification may incorrectly penalize valid dynamic regions or, conversely, fail to detect genuine 3D inconsistencies masked by object motion. As a result, the reward may become noisy or misleading during post-training on scenes with significant dynamic content. A promising future direction is to incorporate motion segmentation to decompose the scene into static background and dynamic foreground, enabling separate consistency evaluation for each component.

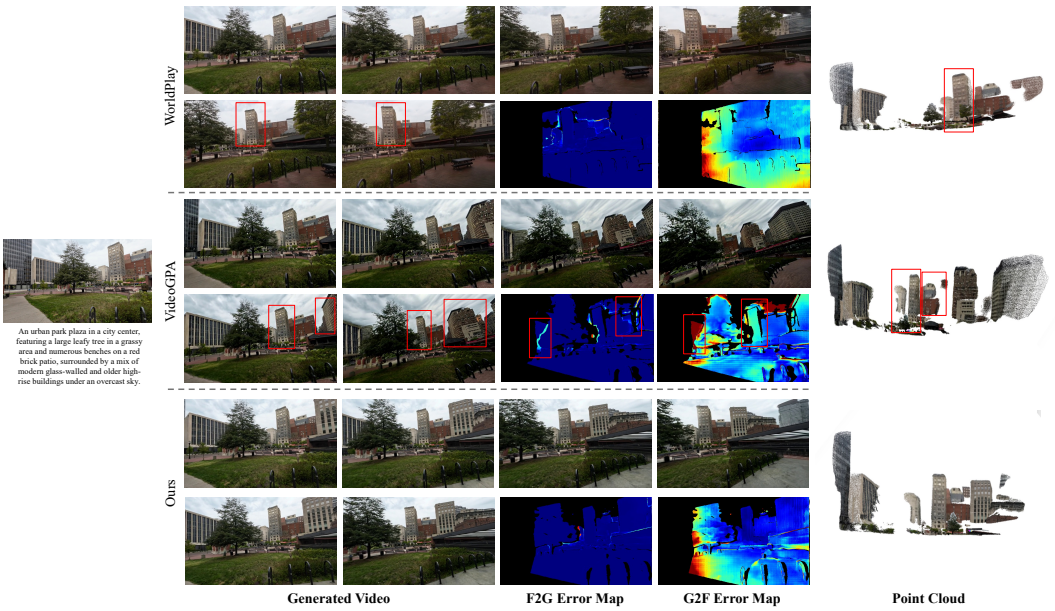

Figure 6: **Qualitative Comparison with VideoGPA and WorldPlay.** We show the generated video, F2E Error Map, G2F Error Map and the reconstructed point cloud.

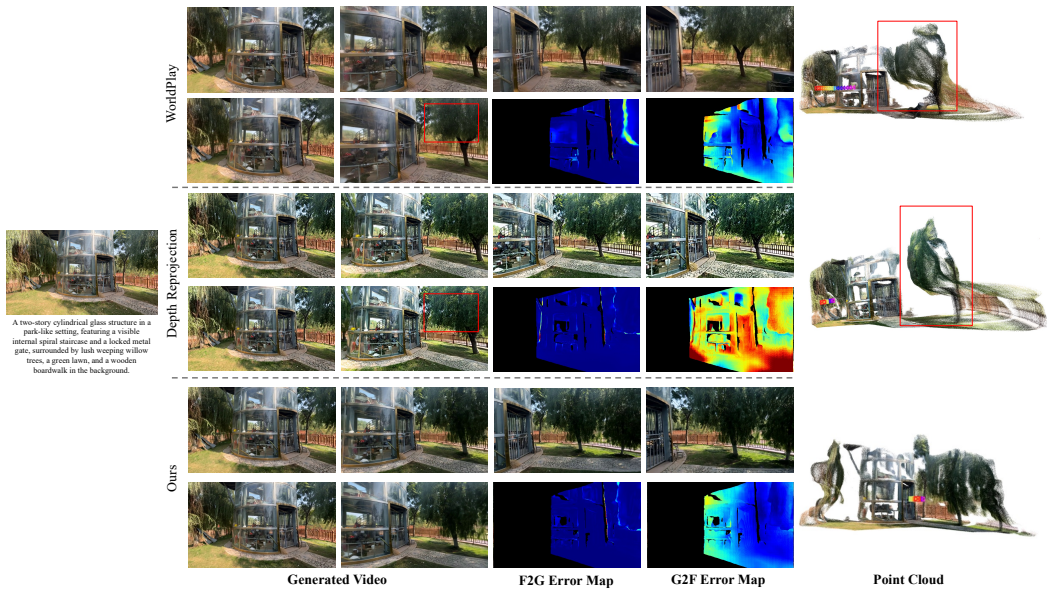

Figure 7: **Qualitative Comparison with Depth Reprojection from VGGRPO and WorldPlay.** We show the generated video, F2E Error Map, G2F Error Map and the reconstructed point cloud.

G Broader Impacts

On the positive side, improving 3D consistency in video world models can benefit robotics and autonomous driving by providing more geometrically faithful simulators, reducing the need for costly real-world data collection, and enabling safer virtual evaluation. On the negative side, advances in realistic video generation could be misused to create more convincing synthetic media for disinformation. However, our method focuses on action-conditioned world models designed for embodied simulation rather than general-purpose video synthesis, thereby limiting the potential for direct misuse. We encourage the community to develop watermarking and detection methods alongside advances in video generation.

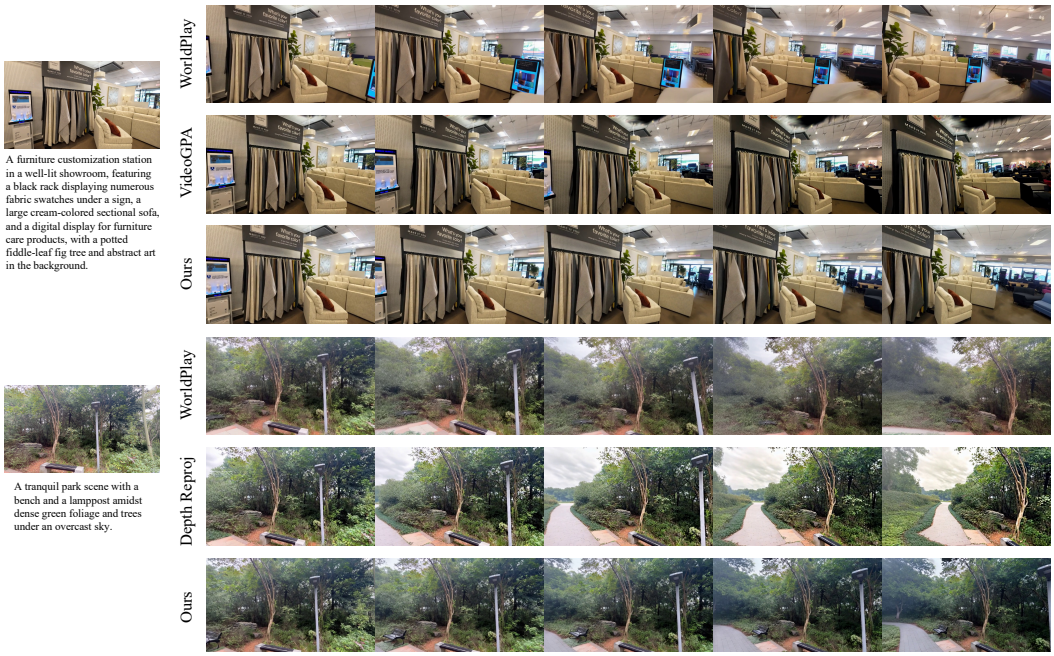

Figure 8: **Visualization of the generated videos.** Compared with other baseline models (WorldPlay, VideoGPA, and Depth Reprojection from VGGRPO), we find that the generated videos by our proposed WORLDPRISM have fewer structure distortions and artifacts, showing better 3D consistency.

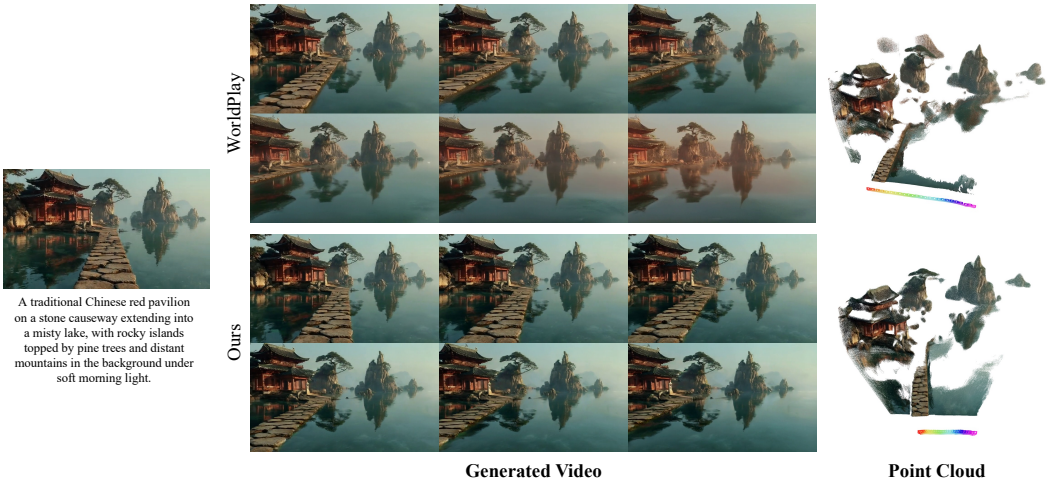

Figure 9: **Visualization of the generated videos.** We show the generated video and the reconstructed point cloud compared with WorldPlay, where our proposed method achieves 3D-consistent world modeling.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The main claims made in the abstract and introduction accurately reflec the paper's contributions and scope.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We discuss the limitatio in the supplementary.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: We provide the full set of assumptions and a complete (and correct) proof.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.

- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The paper fully disclose all the information needed to reproduce the main experimental results of the paper.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: We will opensource the code and data later.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.

- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The paper specify almost all the training and test details.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: The paper does not report error bars.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: For each experiment, the paper provides sufficient information on the computer resources.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: We do.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: We have discussed in our Broader Imapcts section.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper poses no such risks.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: We do.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: The paper does not release new assets.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing nor research with human subjects.

    Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The core method development in this research does not involve LLMs as any important, original, or non-standard components.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.